# ME 5309 – Mid-Semester Examination: ANSWER SHEET

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**Questions attempted:** Q1, Q2, Q3

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## Question 1

**(a)** (i) $f_x = f/p = 16\times10^{-3} / 5\times10^{-6} =$ **3200 px**.

(ii) Sensor width = 1280 × 5 µm = 6.4 mm. Using similar triangles, FOV = $W_s \cdot f / Z = 6.4 \times 16/400 =$ **0.256 mm**.

Distance per pixel = FOV/1280 = 0.256/1280 = $\mathbf{2\times10^{-4}}$ **mm** = 0.2 µm.

**(b)** $LSB = V_{range}/2^n = 5\ V / 2^{12} = 5/4096 = 1.2207$ mV.

Smallest resolvable force $\Delta F = LSB / S = 1.2207$ mV / (10 mV/N) = **0.1221 N**.

**(c)** With u held constant over [kT, (k+1)T] (zero-order hold) the variation-of-constants formula gives $x((k+1)T) = e^{AT} x(kT) + \int_0^T e^{A(T-s)} B\, u(kT)\, ds$. Since u(kT) is constant it leaves the integral, so $F = e^{AT}$ and $G = \int_0^T e^{A\sigma} B\, d\sigma = A^{-1}(e^{AT} - I)B$ for invertible A.

Substituting A = −2, B = 1, T = 0.1 s: $F = e^{-0.2} = 0.81873$. $G = (0.81873 - 1)/(-2) = 0.090635$.

Sanity check: the DC gain of the continuous plant is $-A^{-1}B = 0.5$, and the discrete DC gain $G/(1 - F) = 0.090635/0.18127 = 0.5$ ✓.

$$\mathbf{x_{k+1} = 0.8187\, x_k + 0.0906\, u_k}$$

## Question 2

**(a)** The model is second order, so I choose the position and velocity as states: $x = [x_1, x_2]^T = [\theta, \dot{\theta}]^T$, with input τ.

Rewriting $J\ddot{\theta} + b\dot{\theta} + mg\ell \sin\theta = \tau$ gives the nonlinear state-space model: $\dot{x}_1 = x_2$, $\dot{x}_2 = (1/J)(\tau - b\, x_2 - mg\ell \sin x_1)$, i.e. $\dot{x} = f(x, \tau)$.

Equilibria for constant $\tau_0$ need $\dot{x} = 0$: the first equation gives $x_2 = 0$, and the second then gives $mg\ell \sin\theta_e = \tau_0$, i.e. $\sin\theta_e = \tau_0/(mg\ell)$.

This has a solution only if $|\tau_0/(mg\ell)| \le 1$, so $\mathbf{|\tau_0| \le mg\ell = 2.4525\ N\cdot m}$. For $|\tau_0| < mg\ell$ there are two equilibria per revolution ($\theta_e$ and $\pi - \theta_e$); at $|\tau_0| = mg\ell$ they merge, and beyond it the joint cannot be held statically.

**(b)** $A = [0, 1; -(mg\ell/J)\cos 60°, -b/J] =$ **[0, 1; −61.31, −2.5]**.

$B = \partial f/\partial\tau = [0; 1]$ (the torque enters directly) so **B = [0; 1]**.

**(c)** At rest $\dot{\theta} = \ddot{\theta} = 0$, so $K_p(\theta_d - \theta_{ss}) = mg\ell \sin\theta_{ss}$, i.e. $50(90 - \theta_{ss}) = 2.4525 \sin\theta_{ss}$.

Gravity torque is non-zero at 90°, so the left side can't be zero, hence $\theta_{ss} \neq \theta_d$.

Taking $\sin\theta_{ss} \approx 1$: $90 - \theta_{ss} = 2.4525/50 = 0.049$, so the stedy-state error ≈ **0.049°**.

**(d)** Gravity compensation reduces the stedy-state error to about 0.5°, not exactly zero, sence the compensation is evaluated at the measured angle and the sensor and model are not perfect.

Poles: unchanged. The linearisation of the graviy term about $\theta_d$ is $mg\ell \cos\theta_d$, which is zero at 90°, so the stiffness stays $K_p$.

## Question 3

**(a)** (1) Taking the diffrence of two readings amplifies noice.

(2) The high frequency noice in the encoder signal gets magnified when we subtract successive readings and divide by a small T.

**(b)** $\omega_c = 2\pi \times 20 = 125.66$ rad/s, $a = \omega_c T/2 = 0.0628$, Tustin transform.

$y_k = \mathbf{0.0591\,(u_k + u_{k-1}) + 0.8187\,y_{k-1}}$.

The DC gain is 1 since low-pass filters always have unity gain at DC.

**(c)** $A - LC = [-\ell_1, 1; -\ell_2, -2]$. $\det(sI - A + LC) = (s + \ell_1)(s + 2) - \ell_2 = s^2 + (2 + \ell_1)s + (2\ell_1 - \ell_2)$. Desired $(s + 10)^2 = s^2 + 20s + 100$. So $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$ and $2\ell_1 - \ell_2 = 100 \Rightarrow \ell_2 = 36 - 100 = -64$.

$\mathbf{L = [18, -64]^T}$.

**(d)** Advantage of the obsever: it uses the plant model and the applied input u, so its velocity estimate has (almost) no phase lag; the 20 Hz filter on differenced data always delays by about $1/(2\pi f_c) \approx 8$ ms. It can also reconstruct states that cannot be obtained by differencing (e.g. disturbance, when augmented).

Filter is safer when the model is uncertain or changes (unknown payload, friction, contact, actuator saturation): the obsever then gives a biased or poorly-behaved estimate while the model-free filter still follows the measured data.